

TASK 3

An agentic AI investment system

Task: A US claims dataset of ~30 million insured lives is one input among several. How would it fit into the system, what other datasets would you combine it with, and which capabilities matter most?

Jonas Denck

Candidate · AI Scientist

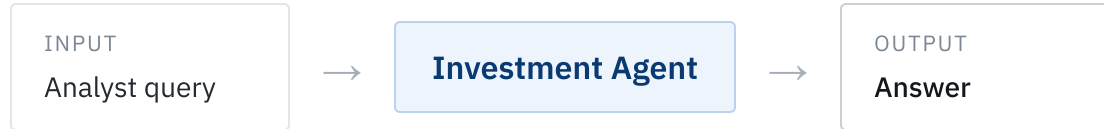
10-minute pitch

AGENDA

01	Simple agent	One analyst, one tool
02	Delegation	Specialist sub-agent
03	Multi-agent ecosystem	Full research coverage
04	Automated	Scheduled + event-driven
05	Uncertainty-aware	Active sensing, human-in-loop

STAGE 1

The simple agent



DATA Claims (IQVIA · MarketScan)

TOOLS TAM estimate

BENEFITS

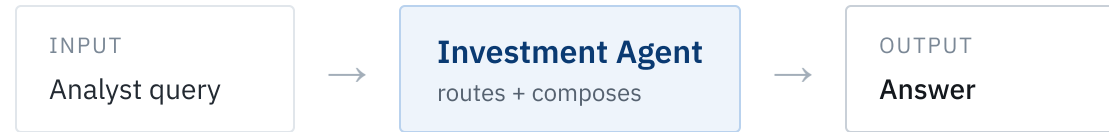
- + Shippable in days; one clear owner
- + Every step inspectable end-to-end

LIMITATIONS

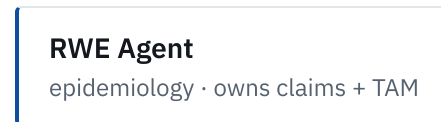
- No specialisation: one agent holds every domain
- Single domain, minimal grounding: hallucination risk
- Analyst must know exactly what to ask
- Limited sense of its own uncertainty, no awareness to reduce it
- Scales only by cloning prompts

STAGE 2

Delegation to a specialist



SUB-AGENT



DATA Claims (IQVIA · MarketScan)

TOOLS TAM estimate

BENEFITS

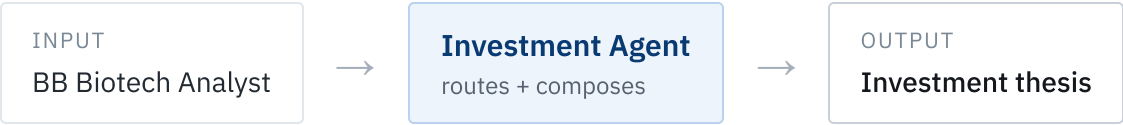
- + Shippable fast; failures localise to one agent
- + Every step inspectable end-to-end
- + Domain expertise isolated, swappable and testable

LIMITATIONS

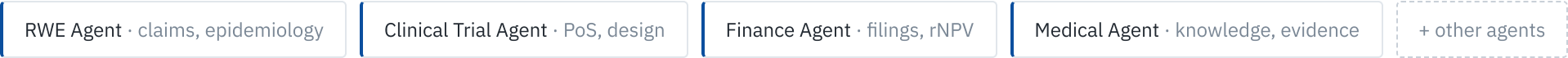
- Single domain, minimal grounding: hallucination risk
- Analyst must know exactly what to ask + scaling limitation
- Limited sense of its own uncertainty, no awareness to reduce it
- Limited scaling, scales mainly by cloning prompts

STAGE 3

Multi-agent ecosystem



SUB-AGENTS



DATA Claims (IQVIA · MarketScan) · EHR (TriNetX · Truveta) · ClinicalTrials.gov · FDA labels · Financials & markets · PubMed · Elicit · Knowledge graph (Open Targets) · Internal document store

TOOLS TAM · hazard models · subpopulation analysis · target–disease association scoring · rNPV · PoS · Monte Carlo simulation · peak-sales forecasting

SKILLS rwd-analysis · rnpv-valuation · financial-modelling · scenario-sensitivity · evidence-grading · thesis-generation · uncertainty-reporting

BENEFITS

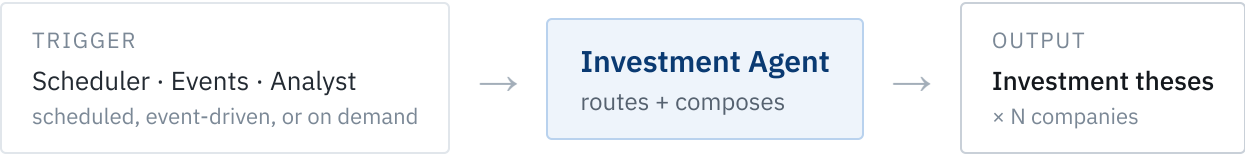
- + Fast to extend; failures stay local
- + Every step inspectable end-to-end
- + Domain expertise isolated and testable
- + Covers a real, grounded thesis across domains

LIMITATIONS

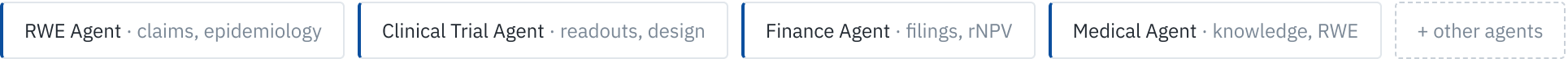
- Analyst must know exactly what to ask + scaling limitation
- Limited sense of its own uncertainty, no awareness to reduce it

STAGE 4

Automated



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BENEFITS

- + Fast to extend; failures stay local
- + Every run inspectable end-to-end
- + Domain expertise isolated and testable
- + Covers a real, grounded thesis across domains
- + Proactive, continuous portfolio coverage

LIMITATIONS

- Limited sense of its own uncertainty, no awareness to reduce it

Uncertainty-aware + active sensing

UNCERTAINTY LAYER

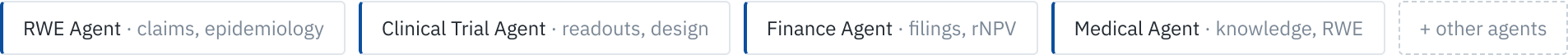
Every output carries an interval, never a point estimate.

ACTIVE SENSING

Identify what would most reduce uncertainty; if that data doesn't exist, ask an investment analyst.



SUB-AGENTS



DATA

Claims (IQVIA · MarketScan) · EHR (TriNetX · Truveta) · ClinicalTrials.gov · FDA labels · Financials & markets · PubMed · Elicit · Knowledge graph (Open Targets) · Internal document store

TOOLS

TAM · hazard models · subpopulation analysis · target–disease association scoring · rNPV · PoS · Monte Carlo simulation · peak-sales forecasting

SKILLS

rwd-analysis · rnpv-valuation · financial-modelling · scenario-sensitivity · evidence-grading · thesis-generation · uncertainty-reporting

BENEFITS

- + Fast to extend; failures stay local
- + Every run inspectable end-to-end
- + Domain expertise isolated and testable
- + Covers a real, grounded thesis across domains
- + Proactive, continuous portfolio coverage
- + Uncertainty awareness: knows what it doesn't know, and asks

LIMITATIONS

- Complexity brings evaluation and maintenance burden

How do we know we built the right system?

1 COMPONENT

Does each tool do its job?

- Unit tests on TAM, rNPV
- Retrieval precision / recall
- Citation resolves to real source

2 TRAJECTORY

Did it reason its way there?

- Correct sub-agent routed
- No skipped or invented steps
- Tool-call argument validity

3 END-TO-END

Is the answer any good?

- Frozen case regression suite
- Analyst preference vs baseline
- Backtest on historical readouts

4 DECISION

Did it change what we did?

- Hit rate on acted-upon calls
- Does the answer change the decision?
- Faster to a confident call

Which capabilities matter most?

Calibration

A stated 70% comes true 70% of the time; usefully wrong beats confidently right.

Uncertainty awareness

Every answer carries an interval, not a bare point estimate.

Grounding

Every claim traces to a verifiable source; no unsupported assertions.